From Prediction to Decision: An Operational Audit Framework for AI Driven Lending, Based on Traditional Banking Controls

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Abstract

AI models are now taking lending decisions that were earlier taken by bank officers. Credit scoring, loan approval and recovery prioritisation are increasingly automated, especially in digital lending. Regulators have responded at the policy level. The Reserve Bank of India has issued Digital Lending Directions in 2025 and draft guidance on model risk management that covers artificial intelligence and machine learning models. However, these frameworks mainly set expectations at the board and policy level. There is limited practical guidance on what oversight of AI lending decisions should look like at the operational level, in daily work. This paper addresses that gap. Drawing on the author's experience of managing 2,500 to 3,000 loan accounts and supporting reporting across 27 branches in a regional rural bank, the paper identifies four traditional banking controls that kept human lending decisions accountable: sanctioning authority chains, credit red flag checks, early warning monitoring of overdue accounts, and maker checker review. It then translates each control into an equivalent operational control for AI lending systems. The paper argues that effective AI oversight in lending does not require inventing new tools. It requires translating controls that banks already understand into the AI context, in a form that small NBFCs and fintech lenders can actually implement. The paper also outlines an open toolkit of templates based on this framework. This extends the author's earlier work on decision centric analytics, which found that systems influence outcomes only when they are integrated into daily decision making.

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JEL Classification: G21, G28, O33

1. Introduction

In a bank branch, every lending decision has a name attached to it.

I know this from experience. I managed a portfolio of 2,500 to 3,000 active loan accounts at a regional rural bank in Gujarat, and supported reporting across 27 branches. When a loan went wrong, the first question was always the same. Who sanctioned it? For a normal loan, the answer was the branch manager. For a big ticket loan, it was a senior manager, the regional manager, or the head office asset department, depending on the amount. The sanctioning record existed. Someone was answerable.

Today, a growing share of lending decisions in India are taken by AI models inside digital lending apps and NBFC systems. The decision happens in seconds. No officer sanctions it. When such a decision goes wrong, the old question, who sanctioned it, has no simple answer. Responsibility is spread across the lender, the app, the model vendor and the data provider.

Regulators have noticed this problem. The Reserve Bank of India issued the Reserve Bank of India (Digital Lending) Directions, 2025 in May 2025, superseding its 2022 guidelines (RBI, 2025a). In August 2024 it released a draft circular on regulatory principles for managing model risks in credit (RBI, 2024). The report of its Committee on Framework for Responsible and Ethical Enablement of Artificial Intelligence, known as FREE AI, followed in August 2025 (RBI, 2025b). In 2026 the RBI issued a broader draft Guidance on Regulatory Principles for Model Risk Management, which explicitly covers third party models and models using artificial intelligence and machine learning (RBI, 2026).

These are important steps. But they mostly work at the board and policy level. They tell the board what policy to approve, what committee should review model validations, and how models should be classified by risk. They say much less about the operational layer. What should the daily exception report of an AI lending system contain? What fields should its audit log capture? Who in a small NBFC should review borderline model decisions, and how?

This operational layer is exactly where traditional banking is strong. Banks have spent decades building daily, practical controls over human lending decisions. This paper argues that these controls can be translated into the AI context, and that this translation gives lenders, especially smaller ones, a practical starting point for AI oversight that they already understand.

The paper builds on my earlier work on decision centric analytics systems (Rajput, 2026), which found that analytics fails not because of weak technology but because it is not integrated into daily decision making. The same logic applies here. AI oversight will fail if it lives only in board policies. It has to live in daily operational routines, the way traditional controls always did.

2. The Problem: What Changes When the Decision Maker Is a Model

Three things change when an AI model replaces a loan officer.

First, accountability becomes diffuse. In the traditional setup, the sanctioning chain was clear and written down. Branch manager for regular loans, higher authorities for larger amounts. In an AI driven setup, the lender may say the vendor built the model, the vendor may say the lender chose the cutoffs, and the data provider may say it only supplied inputs. No single desk owns the decision.

Second, scale outruns review capacity. A branch manager could look again at any file on their desk. A model scoring thousands of applications a day cannot be reviewed file by file. Oversight has to become selective and systematic, which is a different skill from case by case review.

Third, explanation becomes harder, at exactly the moment it is needed most. In my branch work, I dealt with customers who could not read their own loan statements. We would sit with them and explain. An explainability requirement written for such a borrower cannot assume they will read a technical reason code. The explanation has to be simple enough for a front line staff member to give verbally. Most model explanation methods today are built for data scientists, not for a branch officer talking to a farmer.

None of these three problems is solved by policy documents alone. All three were once solved, for human decisions, by operational controls. The next section describes those controls as they actually worked.

3. What Banks Already Do: Four Controls Over Human Decisions

This section is based on direct experience in regional rural banking in India between 2017 and 2022, and in retail banking before that. I describe four controls in the plain form in which they operated.

3.1 Sanctioning authority chains

Every loan had a sanctioning authority based on the ticket size. Regular loans were sanctioned by the branch manager. Larger amounts moved up to the senior manager, the regional manager, or the head office asset department. The sanction record named the authority. When an account later turned bad, review started from that record. The control here is simple. Authority is tied to stake, and the decision is attributable to a named role.

3.2 Credit red flag checks before sanction

Before sanction, standard red flags were checked. These included existing overdues, low savings account balance, a low credit bureau score, other pending loans, and red flags in the CIBIL report. These checks were not advanced analytics. They were a fixed list, applied every time, and a flagged case needed extra justification or a higher authority to proceed. The control is a standard pre decision checklist with an escalation rule.

3.3 Early warning monitoring of accounts

After disbursement, accounts were monitored continuously. The first trigger was simple. One EMI overdue put the account on a watch list. The head office also circulated NPA lists that branches had to act on. The value of this control was its timing. Problems were surfaced at one missed payment, not after the account was already deep in default. The control is outcome monitoring with a defined early trigger and a defined owner for follow up.

3.4 Maker checker review

For sensitive daily operations, two people were involved by design. Cash debit and credit entries and cheque passing required one person to make the entry and another to check and authorise it. Neither could complete the transaction alone. The control is that no high stakes action is completed by a single actor.

These four controls share one feature. They are boring, daily and specific. That is why they worked. They did not depend on anyone's judgement about whether oversight was needed today. They ran every day by default.

4. The Translation: Four Equivalent Controls for AI Lending Systems

This section maps each traditional control to an operational control for AI lending. For each, I state the AI equivalent, what it requires in practice, and what does not translate cleanly.

4.1 Sanctioning authority chain becomes a decision ownership register

AI equivalent. Every model driven decision type is assigned to a named human role, in writing, before deployment. For example, the credit head owns approval and rejection decisions of the personal loan model, version 3.2, deployed on a stated date. The register records the model version, the decision types it takes, the cutoffs chosen, who approved those cutoffs, and who is answerable for outcomes.

In practice. This is one document, reviewed whenever the model or cutoffs change. It restores the old question, who sanctioned it, in a new form. The answer becomes, this model version, with these cutoffs, approved by this named role on this date.

What does not translate. In the old system, the sanctioning authority personally reviewed the file. The owner of a model cannot review each decision. Ownership here means answerability for the system and its settings, not for each individual file. This is a real weakening and it should be stated honestly, not hidden.

4.2 Red flag checks become model input and override rules

AI equivalent. The fixed pre sanction checklist becomes two things. First, a documented list of what the model is allowed to use as input, and what it must not use. Second, a set of hard rules that sit outside the model. For example, a bureau level fraud flag always stops auto approval regardless of the model score, and such a case routes to a human.

In practice. The old checklist logic survives as guardrails around the model. The lender writes down which conditions can never be overridden by a good model score. This is a one page policy, not a technical system.

What does not translate. The old checklist was fully visible to the officer using it. A model's internal weighting is not visible in the same way. The guardrail list controls the boundary of the model's authority, but it does not make the inside of the model transparent. Transparency of the model itself is a separate and harder problem.

4.3 Early warning monitoring becomes outcome monitoring by segment

AI equivalent. The one EMI overdue trigger already exists in lending operations. What changes is the reporting cut. Overdue and default outcomes are tracked by model version and by borrower segment, monthly. If accounts approved by one model version in one segment, for example new to credit borrowers in a particular region, start going overdue faster than others, that pattern is the new early warning. It signals that the model may be scoring that segment wrongly.

In practice. This is a standard dashboard. Rows are borrower segments, columns are model versions, values are early overdue rates. Any bank analyst who has built NPA reporting can build it. The head office NPA list has a direct descendant here, a monthly model outcome report circulated with named owners for follow up.

What does not translate. Outcome monitoring is slow by nature. A bad scoring pattern shows up only after some months of repayment behaviour. It catches drift, not immediate individual errors. It must be paired with the guardrails of Section 4.2, which act before the decision.

4.4 Maker checker becomes a human review tier for borderline and high stakes decisions

AI equivalent. The model alone does not complete every decision. Decisions are split into three bands. Clear approvals and clear rejections may complete automatically. Borderline scores, high ticket amounts, and cases with guardrail flags route to a human checker before completion. The band boundaries are set by the same authority that owns the model under Section 4.1.

In practice. This is queue design. Most digital lenders already have manual review queues for fraud. The change is to define the queue by policy, record the checker's decision against the

model's recommendation, and review disagreement patterns monthly. Where checkers frequently overturn the model in one segment, that segment's cutoffs need review.

What does not translate. In cheque passing, maker and checker had equal visibility of the instrument. Here the checker sees the application but not the model's full reasoning. The checker is checking the outcome against their judgement, not repeating the model's work. Checker fatigue and rubber stamping are known risks, and the monthly disagreement review exists partly to detect exactly that.

4.5 The framework in one view

Traditional control	AI equivalent	Core artefact
Sanctioning authority chain	Decision ownership register	One register document
Pre sanction red flag checklist	Input rules and hard guardrails	One page guardrail policy
One EMI overdue watch list, NPA list	Outcome monitoring by model version and segment	Monthly dashboard and report
Maker checker on cash and cheques	Human review tier for borderline and high stakes cases	Review queue with logged decisions

5. Why This Matters Now in India

Two regulatory developments make this framework timely.

First, the Reserve Bank of India (Digital Lending) Directions, 2025, issued in May 2025, consolidated and strengthened the rules for digital lenders, superseding the 2022 guidelines, with emphasis on consumer protection, disclosure and accountability in arrangements involving third party fintech providers (RBI, 2025a).

Second, the RBI has moved on model risk itself. A draft circular on regulatory principles for management of model risks in credit was issued in August 2024 (RBI, 2024). The FREE AI committee report on responsible and ethical enablement of artificial intelligence in the financial sector followed in August 2025 (RBI, 2025b). A broader draft Guidance on Regulatory Principles for Model Risk Management, covering third party models and artificial intelligence and machine learning models, was issued for public comments in 2026 (RBI, 2026). Commentary on the draft indicates a risk based tiering of models by materiality, complexity, consumer impact and explainability.

These documents answer the question of what boards and risk committees must do. The framework in this paper answers a different question. What should the compliance officer of a mid sized NBFC actually set up on Monday morning? The four artefacts in Table 4.5, a register, a guardrail page, a monthly dashboard and a review queue, are deliberately modest. A lender that cannot afford a model validation team can still maintain all four. In a market where regulatory capacity and firm level compliance capacity are both stretched, modest and implementable beats sophisticated and ignored.

There is also a practical alignment benefit. If the final RBI framework requires model tiering and board accountability, a lender that already runs these four operational controls will find

compliance far easier, because the controls generate exactly the records that supervision will ask for.

6. An Open Toolkit

To make the framework usable, I am releasing template versions of the four artefacts in my public research repository on GitHub:

1. A decision ownership register template, with suggested fields.

2. A one page guardrail policy template for AI lending decisions.

3. An outcome monitoring dashboard template, buildable in Power BI or similar tools, with the segment by model version structure described in Section 4.3.

4. An exception and review queue log template, recording model recommendation, checker decision and reason.

The templates are starting points, not standards. Lenders should adapt them to their products and scale. The repository will also track revisions as feedback comes in.

7. Limitations and Open Questions

This framework has clear limits, and stating them is part of the argument.

First, it does not solve model transparency. The controls here manage the boundary and the outcomes of a model. They do not open the model itself. Explainability for low literacy borrowers, raised in Section 2, remains an open problem that this framework only routes around, by keeping humans in the loop for borderline cases.

Second, it is based on experience in one country's banking system. The four traditional controls are common in Indian banking, and versions of them exist elsewhere, but the translation may differ in other regulatory environments.

Third, the framework has not yet been tested end to end in a live digital lender. It is a practitioner's translation, offered for adoption and criticism. Evidence from actual implementation, especially on whether the human review tier avoids rubber stamping, is the most important next step.

Fourth, this paper deliberately does not address fairness testing, bias audits or data protection compliance in depth. These matter, and they interact with the controls here, especially outcome monitoring by segment, which can surface disparate outcomes. A fuller treatment is left for future work.

8. Conclusion

Banking already knows how to keep decisions accountable. It has done so for decades with unglamorous tools. A named sanctioning authority. A fixed checklist of red flags. A watch list that starts at one missed EMI. Two people on every cheque.

AI lending broke the link between decisions and these controls, because the controls were designed around human decision makers. This paper's argument is that the link can be rebuilt. Each control has a workable AI equivalent, and together they form an operational oversight layer that sits under the policy level frameworks regulators are now building.

My earlier work found that analytics changes outcomes only when it enters daily decision making. Oversight is no different. If AI governance in lending remains a board document, it will fail quietly. If it becomes a register, a guardrail page, a monthly report and a review queue, it has a chance of working, because that is the shape in which banking controls have always actually worked.

Declaration of Interest

The author declares no conflict of interest.

Ethics Approval

This study does not involve human participants or sensitive personal data and does not require ethical approval. All descriptions of banking practice are general and do not identify any individual or institution beyond the author's own stated work history.

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